RNASeq-Pasilla

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Introduction

This analysis reproduces, in part, the workflow and findings of a 2010 study investigating the *Pasilla* gene using RNA-seq data (reference (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3032923/)). In the original study, *Pasilla* expression was experimentally depleted prior to RNA isolation, and both single-end and paired-end sequencing libraries were prepared for treated and untreated samples. The resulting RNA-seq libraries were sequenced and analyzed to assess differential gene expression.

In this script, we compare treated versus untreated samples to evaluate the transcriptional effects of *Pasilla* depletion, while accounting for sequencing strategy (single-end vs. paired-end) as an additional experimental factor.

```
knitr::opts_chunk$set(echo = TRUE)

# Load required libraries
library(DESeq2)
```

```
## Loading required package: S4Vectors
```

```
## Loading required package: stats4
```

```
## Loading required package: BiocGenerics
```

```
## Loading required package: generics
```

```
##
## Attaching package: 'generics'
```

```
## The following objects are masked from 'package:base':
##
##     as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
##     setequal, union
```

```
##
## Attaching package: 'BiocGenerics'
```

```
## The following objects are masked from 'package:stats':
##
##     IQR, mad, sd, var, xtabs
```

```
## The following objects are masked from 'package:base':
##
##     anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##     colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##     get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
##     order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
##     rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
##     unsplit, which.max, which.min
```

```
##
## Attaching package: 'S4Vectors'
```

```
## The following object is masked from 'package:utils':
##
##     findMatches
```

```
## The following objects are masked from 'package:base':
##
##     expand.grid, I, unname
```

```
## Loading required package: IRanges
```

```
##
## Attaching package: 'IRanges'
```

```
## The following object is masked from 'package:grDevices':
##
##     windows
```

```
## Loading required package: GenomicRanges
```

```
## Loading required package: Seqinfo
```

```
## Loading required package: SummarizedExperiment
```

```
## Loading required package: MatrixGenerics
```

```
## Loading required package: matrixStats
```

```
##
## Attaching package: 'MatrixGenerics'
```

```
## The following objects are masked from 'package:matrixStats':
##
##     colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
##     colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
##     colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
##     colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
##     colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
##     colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
##     colWeightedMeans, colWeightedMedians, colWeightedSds,
##     colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
##     rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
##     rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
##     rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
##     rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
##     rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
##     rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
##     rowWeightedSds, rowWeightedVars
```

```
## Loading required package: Biobase
```

```
## Welcome to Bioconductor
##
##     Vignettes contain introductory material; view with
##     'browseVignettes()'. To cite Bioconductor, see
##     'citation("Biobase")', and for packages 'citation("pkgname")'.
```

```
##
## Attaching package: 'Biobase'
```

```
## The following object is masked from 'package:MatrixGenerics':
##
##     rowMedians
```

```
## The following objects are masked from 'package:matrixStats':
##
##     anyMissing, rowMedians
```

```
library(pheatmap)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following object is masked from 'package:Biobase':
##
##     combine
```

```
## The following object is masked from 'package:matrixStats':
##
##     count
```

```
## The following objects are masked from 'package:GenomicRanges':
##
##     intersect, setdiff, union
```

```
## The following object is masked from 'package:Seqinfo':
##
##     intersect
```

```
## The following objects are masked from 'package:IRanges':
##
##     collapse, desc, intersect, setdiff, slice, union
```

```
## The following objects are masked from 'package:S4Vectors':
##
##     first, intersect, rename, setdiff, setequal, union
```

```
## The following objects are masked from 'package:BiocGenerics':
##
##     combine, intersect, setdiff, setequal, union
```

```
## The following object is masked from 'package:generics':
##
##     explain
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(RColorBrewer)
library(ggplot2)
library(ggrepel)
library(clusterProfiler)
```

```
##
```

```
## clusterProfiler v4.18.3 Learn more at https://yulab-smu.top/contribution-knowledge-mining/
##
## Please cite:
##
## T Wu, E Hu, S Xu, M Chen, P Guo, Z Dai, T Feng, L Zhou, W Tang, L Zhan,
## X Fu, S Liu, X Bo, and G Yu. clusterProfiler 4.0: A universal
## enrichment tool for interpreting omics data. The Innovation. 2021,
## 2(3):100141
```

```
##
## Attaching package: 'clusterProfiler'
```

```
## The following object is masked from 'package:IRanges':
##
##     slice
```

```
## The following object is masked from 'package:S4Vectors':
##
##     rename
```

```
## The following object is masked from 'package:stats':
##
##     filter
```

```
library(goseq)
```

```
## Loading required package: BiasedUrn
```

```
## Loading required package: geneLenDataBase
```

```
##
## Attaching package: 'geneLenDataBase'
```

```
## The following object is masked from 'package:S4Vectors':
##
##     unfactor
```

Data Setup

The dataset is organized into treated and untreated samples, with metadata describing sequencing strategy and treatment status. These factors allow us to isolate the effects of *Pasilla* depletion while controlling for technical variation introduced by different sequencing protocols.

```r
# Read in raw count matrix

count_data <- read.csv("data/count_matrix.csv", header = TRUE, row.names = 1)

# Read in sample metadata

sample_info <- read.csv("data/design.csv", header = TRUE, row.names = 1)

# Inspect sample metadata

head(sample_info)
```

```
##                             Treatment Sequencing
## GSM461176_untreated_single untreated         SE
## GSM461177_untreated_paired untreated         PE
## GSM461178_untreated_paired untreated         PE
## GSM461179_treated_single     treated         SE
## GSM461180_treated_paired     treated         PE
## GSM461181_treated_paired     treated         PE
```

```r
# Define experimental factors

sample_info$Treatment <- factor(sample_info$Treatment)
sample_info$Sequencing <- factor(sample_info$Sequencing)
```

DESeq2 Setup

Here, we construct the DESeq2 dataset object and specify the experimental design. Sequencing strategy is included as a covariate to account for systematic differences between single-end and paired-end libraries. The untreated samples are explicitly set as the reference level for treatment comparisons.

```r
# Create DESeq2 dataset

dds <- DESeqDataSetFromMatrix(
countData = count_data,
colData = sample_info,
design = ~ Sequencing + Treatment
)

# Set untreated samples as the reference level

dds$Treatment <- factor(dds$Treatment, level = c("untreated", "treated"))
```

Data Filtering and Differential Expression Analysis

To improve statistical robustness, genes with consistently low counts are removed prior to analysis. Specifically, we retain genes with counts greater than 10 in at least as many samples as the smallest treatment group. Differential expression analysis is then performed using DESeq2, and results are ranked by statistical significance.

```r
# Filter out lowly expressed genes

keep <- rowSums(counts(dds) > 10) >= min(table(sample_info$Treatment))
dds <- dds[keep, ]

# Run differential expression analysis

dds <- DESeq(dds)
```

```
## estimating size factors
```

```
## estimating dispersions
```

```
## gene-wise dispersion estimates
```

```
## mean-dispersion relationship
```

```
## final dispersion estimates
```

```
## fitting model and testing
```

```r
deseq_results <- results(dds)

# Convert results to data frame and sort by p-value

deseq_results <- as.data.frame(deseq_results)
deseq_results <- deseq_results[order(deseq_results$pvalue), ]

# Inspect top differentially expressed genes

head(deseq_results)
```

```
##               baseMean log2FoldChange      lfcSE      stat        pvalue
## FBgn0039155  1087.088      -4.599523 0.15245724 -30.16927 5.994855e-200
## FBgn0003360  6410.646      -3.164152 0.10938392 -28.92703 5.459701e-184
## FBgn0026562 65125.951      -2.465742 0.08754260 -28.16619 1.518038e-174
## FBgn0025111  2194.660       2.831895 0.10143424  27.91853 1.589601e-171
## FBgn0029167  5432.388      -2.190908 0.09756613 -22.45562 1.127744e-111
## FBgn0035085   928.391      -2.584359 0.12147633 -21.27459 1.952308e-100
##                      padj
## FBgn0039155 4.997910e-196
## FBgn0003360 2.275877e-180
## FBgn0026562 4.218629e-171
## FBgn0025111 3.313126e-168
## FBgn0029167 1.880401e-108
## FBgn0035085  2.712733e-97
```

Examination of Specific Genes

We next examine individual genes of interest to confirm expected biological effects. In particular, the *Pasilla* gene is expected to be downregulated in treated samples. This is supported by a negative log2 fold change and a statistically significant p-value.

```
# Inspect a specific gene of interest

deseq_results["FBgn0039155", ]
```

```
##             baseMean log2FoldChange     lfcSE      stat        pvalue
## FBgn0039155 1087.088      -4.599523 0.1524572 -30.16927 5.994855e-200
##                     padj
## FBgn0039155 4.99791e-196
```

```
# Evaluate Pasilla gene expression changes

deseq_results["FBgn0261552", ]
```

```
##             baseMean log2FoldChange     lfcSE      stat       pvalue
## FBgn0261552 8915.949      -1.780947 0.1510969 -11.78679 4.566333e-32
##                     padj
## FBgn0261552 1.312742e-29
```

Identification of Differentially Expressed Genes

Applying significance and effect-size thresholds (adjusted p-value < 0.05 and |log2 fold change| > 1), we identify a refined set of differentially expressed genes suitable for downstream analysis and visualization.

```
# Select statistically significant and biologically meaningful genes

filtered <- deseq_results %>%
filter(padj < 0.05) %>%
filter(abs(log2FoldChange) > 1)

# Number of differentially expressed genes

dim(filtered)
```

```
## [1] 234   6
```

Storing Results

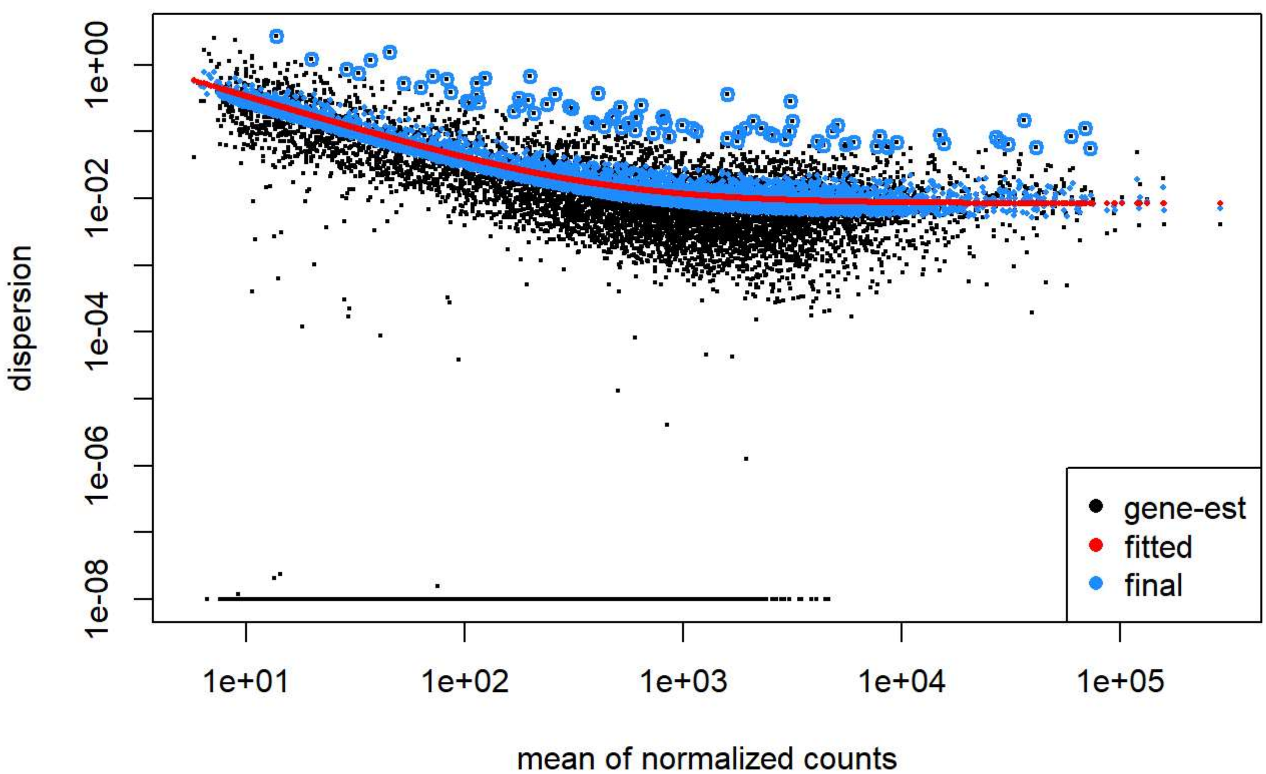
Key results, including full DESeq2 outputs, filtered gene lists, and normalized counts, are written to disk to facilitate reproducibility and downstream analyses.

```
# Save DESeq2 results

write.csv(deseq_results, "data/deseq_result.all.csv")
write.csv(filtered, "data/deseq_result.filtered.csv")

# Export normalized counts

normalized_counts <- counts(dds, normalized = TRUE)
write.csv(normalized_counts, "data/normalized_counts.csv")
```

Visualization: Dispersion and Principal Component Analysis

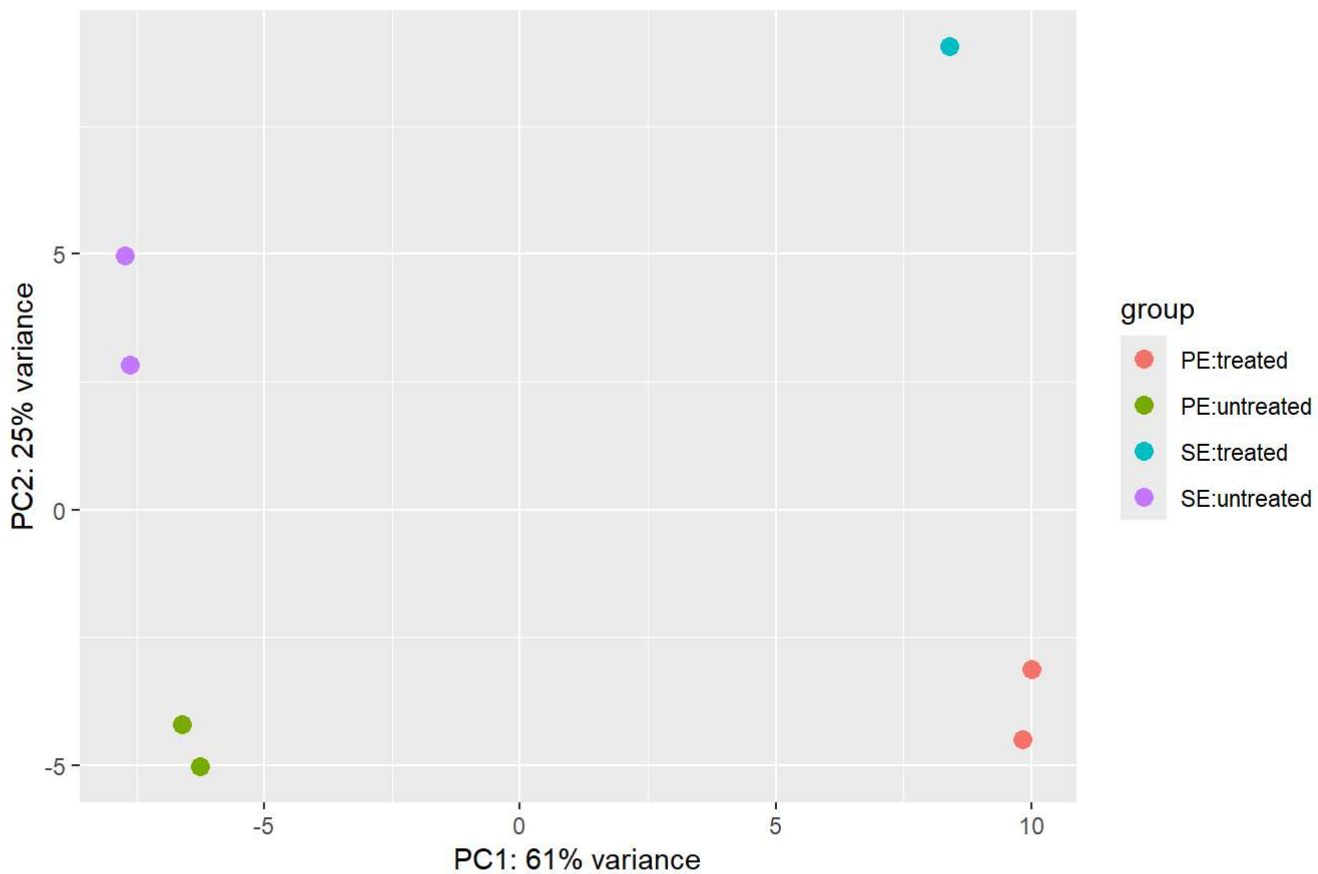
Dispersion estimates are examined to assess model fit, followed by principal component analysis (PCA) to evaluate global patterns in the data. Clear separation between treated and untreated samples indicates a strong treatment effect, while additional structure reflects differences between sequencing strategies.

```
# Dispersion plot

plotDispEsts(dds)
```



```r
# Variance-stabilizing transformation

vsd <- vst(dds, blind = FALSE)

# PCA plot

plotPCA(vsd, intgroup = c("Sequencing", "Treatment"))
```

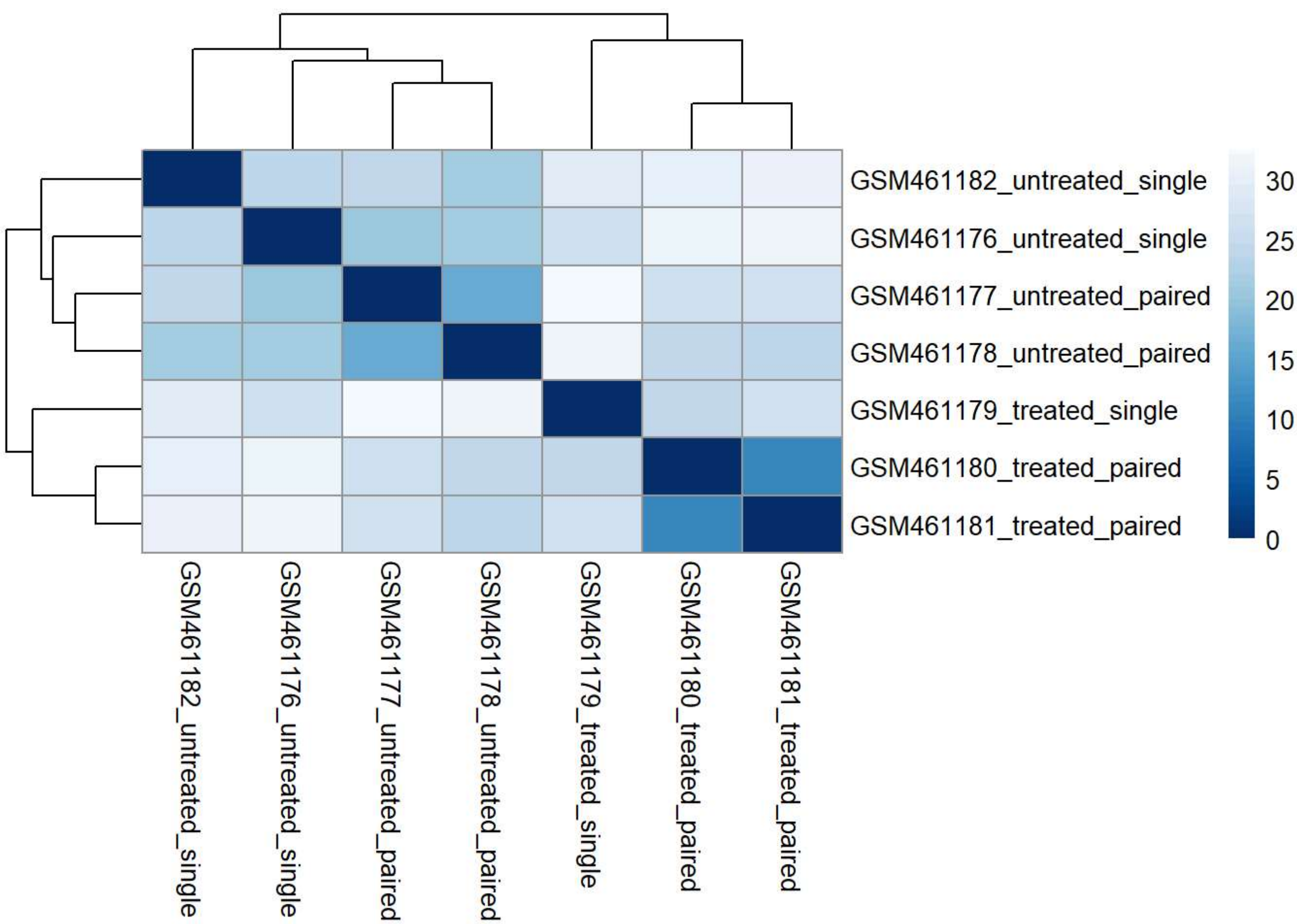
```
## using ntop=500 top features by variance
```



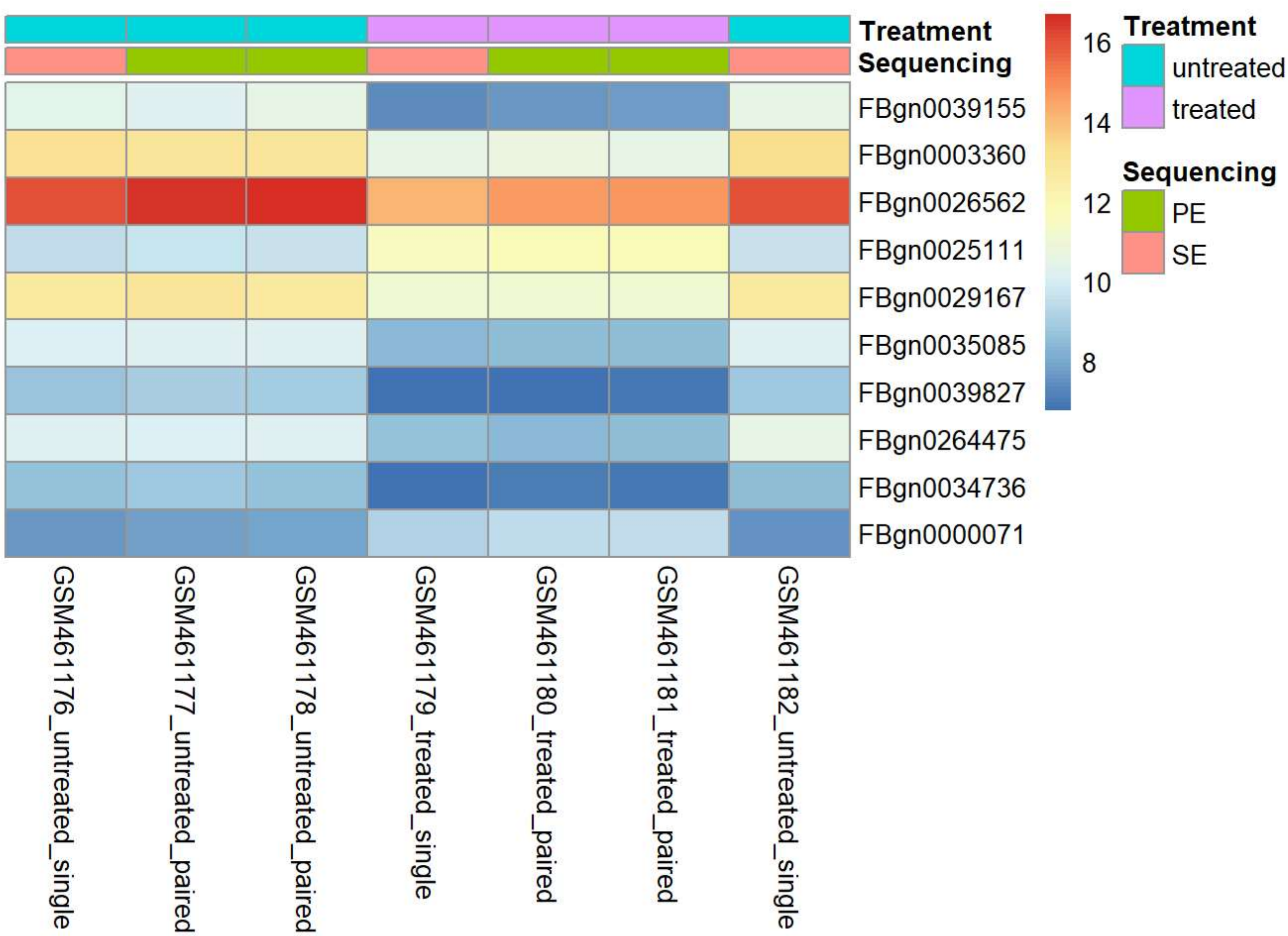
Visualization: Heatmaps and Differential Expression Summaries

Sample-to-sample distance heatmaps highlight similarities and differences among libraries. Additional heatmaps focusing on top differentially expressed genes provide a more detailed view of expression patterns across conditions.

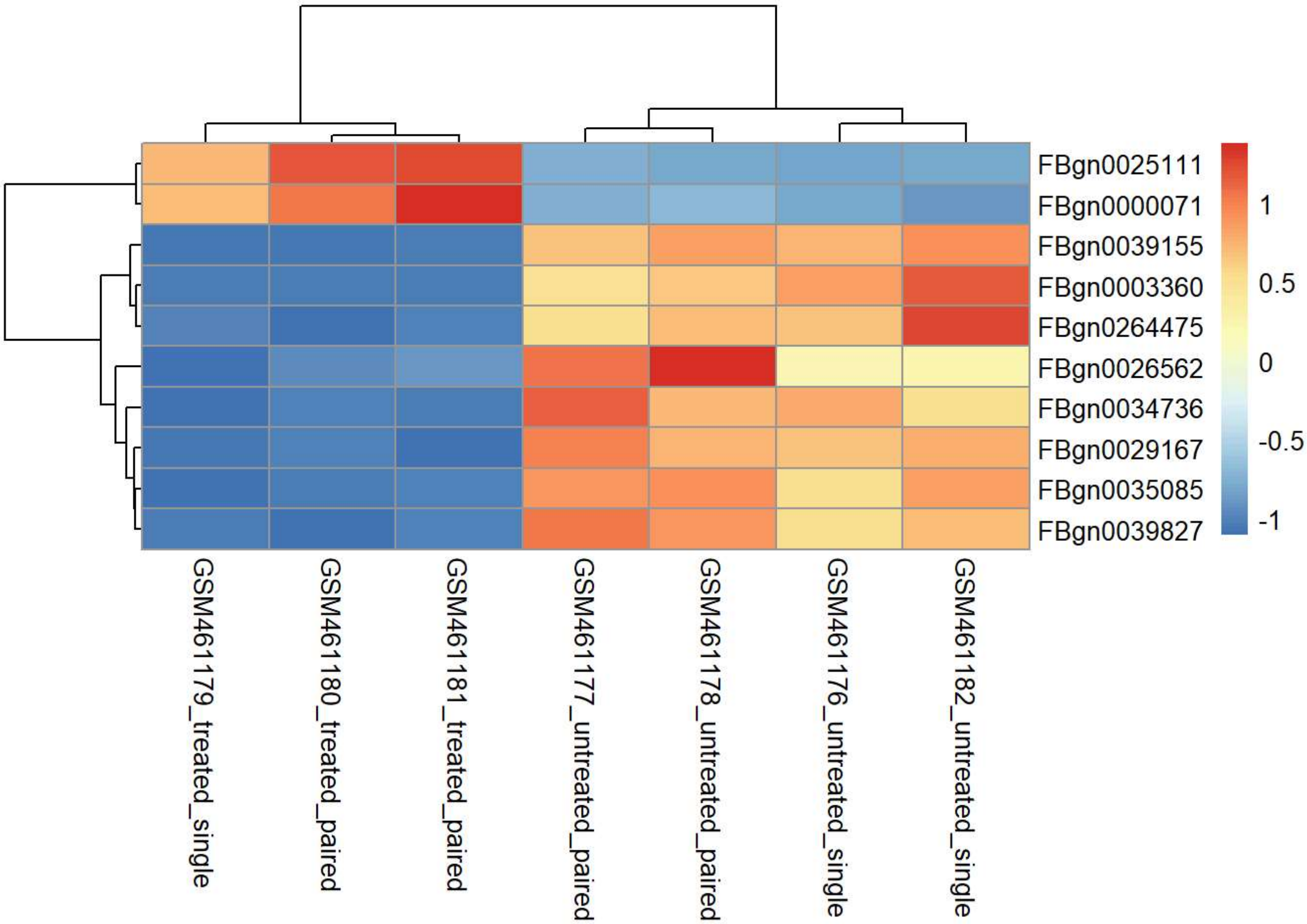
```
# Compute sample distance matrix

sample_dists <- dist(t(assay(vsd)))
sample_dists_matrix <- as.matrix(sample_dists)

# Define color palette

colors <- colorRampPalette(rev(brewer.pal(9, "Blues")))(255)

# Sample distance heatmap

pheatmap(
sample_dists_matrix,
clustering_distance_rows = sample_dists,
clustering_distance_cols = sample_dists,
color = colors
)
```



```
# Identify top 10 genes by adjusted p-value

top_hits <- row.names(deseq_results[order(deseq_results$padj), ][1:10, ])
write.csv(top_hits, "data/top_hits.csv")

# Regularized log transformation

rld <- rlog(dds, blind = FALSE)

# Annotated heatmap of top hits

annot_info <- as.data.frame(colData(dds)[, c("Sequencing", "Treatment")])
pheatmap(
assay(rld)[top_hits, ],
cluster_rows = FALSE,
cluster_cols = FALSE,
show_rownames = TRUE,
annotation_col = annot_info
)
```



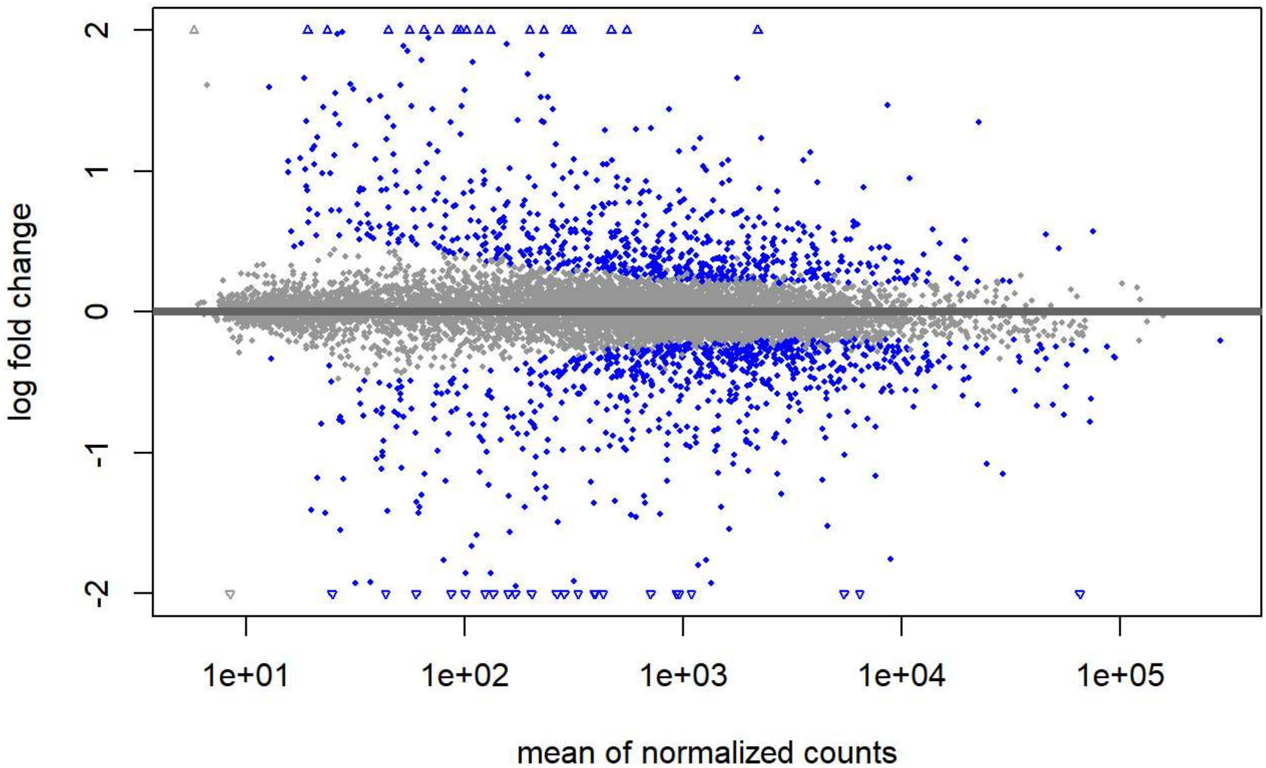
```r
# Z-score heatmap

cal_z_score <- function(x) (x - mean(x)) / sd(x)
z_scores <- t(apply(normalized_counts, 1, cal_z_score))
pheatmap(z_scores[top_hits, ])
```



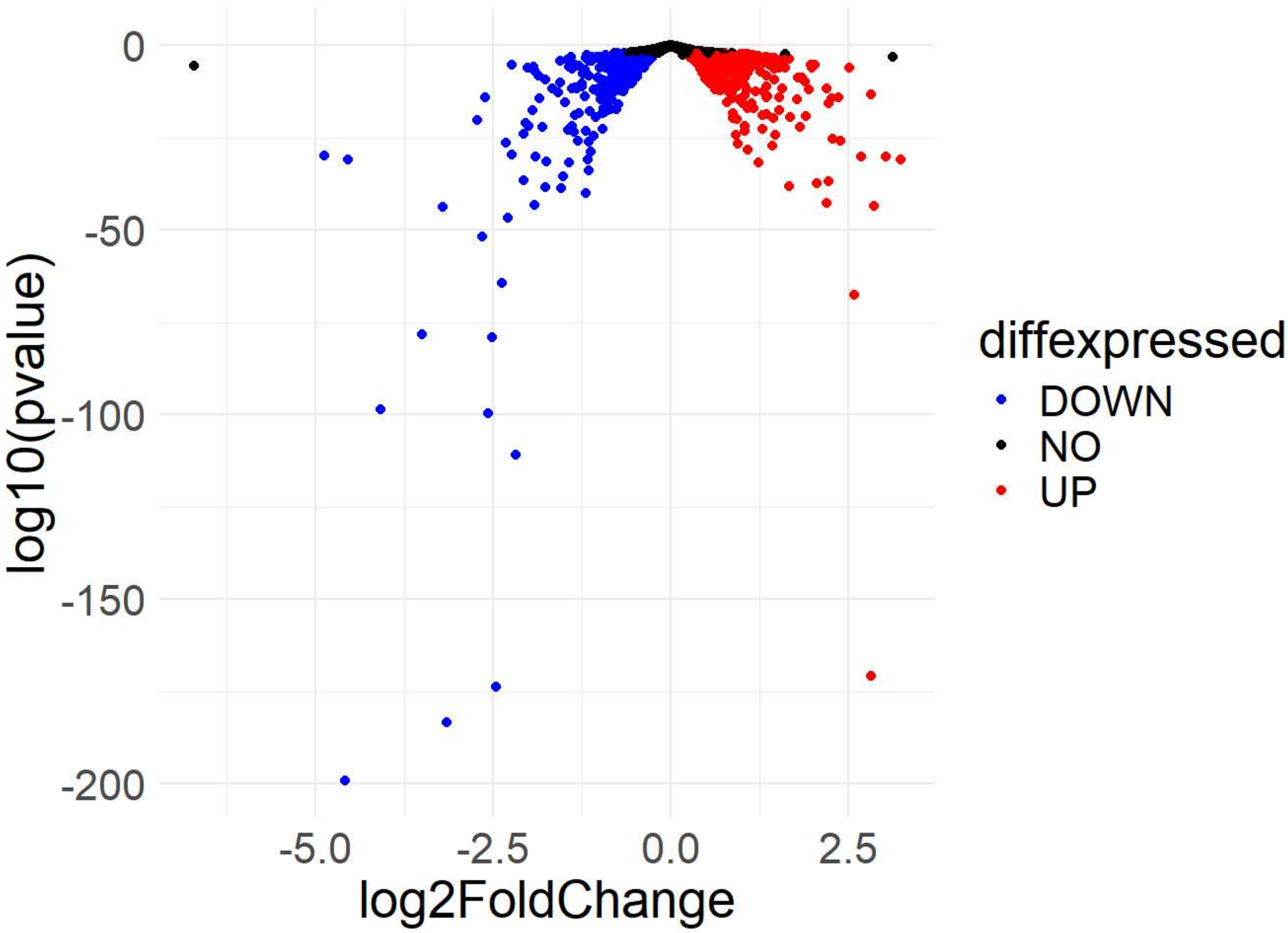

```r
# MA plot with log-fold change shrinkage

resLFC <- lfcShrink(dds, coef = "Treatment_treated_vs_untreated", type = "apeglm")
```

```
## using 'apeglm' for LFC shrinkage. If used in published research, please cite:
##     Zhu, A., Ibrahim, J.G., Love, M.I. (2018) Heavy-tailed prior distributions for
##     sequence count data: removing the noise and preserving large differences.
##     Bioinformatics. https://doi.org/10.1093/bioinformatics/bty895
```

```r
plotMA(resLFC, ylim = c(-2, 2))
```




```r
# Volcano plot

resLFC <- as.data.frame(resLFC)
resLFC$diffexpressed <- "NO"
resLFC$diffexpressed[resLFC$log2FoldChange > 0.1 & resLFC$padj < 0.05] <- "UP"
resLFC$diffexpressed[resLFC$log2FoldChange < -0.1 & resLFC$padj < 0.05] <- "DOWN"

# Generate volcano plot

ggplot(resLFC, aes(x = log2FoldChange, y = log10(pvalue), color = diffexpressed)) +
geom_point() +
theme_minimal() +
scale_color_manual(values = c("blue", "black", "red")) +
theme(text = element_text(size = 20))
```



Gene Ontology (GO) Analysis

Gene Ontology enrichment was run externally using the Galaxy platform. Overall, 60 GO terms (about 0.5%) were significantly over-represented, while 7 terms (about 0.07%) were under-represented. Most of the over-represented terms fell under Biological Processes, with a smaller number related to Cellular Components and Molecular Functions. The under-represented terms were mostly Biological Processes, with a couple of Cellular Component terms and no Molecular Function terms.

KEGG Pathway Analysis

The KEGG pathway analysis showed that roughly 2% of pathways were significantly over-represented. In particular, the general metabolic pathway (01100) and the Glycolysis/Gluconeogenesis pathway (00010) stood out, pointing to broad metabolic changes linked to Pasilla depletion. No pathways were significantly under-represented. Visual summaries of these pathways are included in the data directory.

Future Directions

This project was a useful hands-on exercise in RNA-seq analysis, including data processing, filtering, visualization, and interpretation using DESeq2. While clear differential expression patterns emerged, future work could focus more on interpreting specific genes and pathways to better tie the statistical results to biological meaning.

It would also be valuable to repeat similar analyses using UNIX-based workflows or Python tools, both to build familiarity with alternative approaches and to compare results across different analysis frameworks.

Sources

- https://academic.oup.com/bioinformatics/article/32/19/3047/2196507 (https://academic.oup.com/bioinformatics/article/32/19/3047/2196507)
- https://training.galaxyproject.org/training-material/topics/transcriptomics/tutorials/goenrichment/tutorial.html (https://training.galaxyproject.org/training-material/topics/transcriptomics/tutorials/goenrichment/tutorial.html)
- https://training.galaxyproject.org/training-material/topics/transcriptomics/tutorials/ref-based/tutorial.html#conclusion (https://training.galaxyproject.org/training-material/topics/transcriptomics/tutorials/ref-based/tutorial.html#conclusion)
- https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3032923/ (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3032923/)
- https://www.genome.jp/kegg/ (https://www.genome.jp/kegg/)